

Automated Question Bank Generation System

AI-powered educational content processing and question-paper generation platform



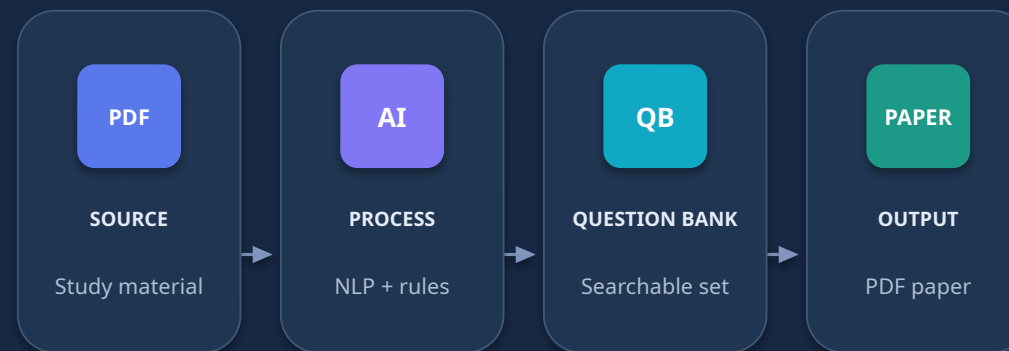
Turn teaching material into a reusable, searchable assessment resource.

TEAM MEMBERS · Name 1 · Name 2 · Name 3

College / Institution: _____
Department: _____

ACADEMIC YEAR 2026-27

FROM SOURCE MATERIAL TO PRINT-READY ASSESSMENT



TEACHER CONTROL

Search · Filter · Edit · Build · Export

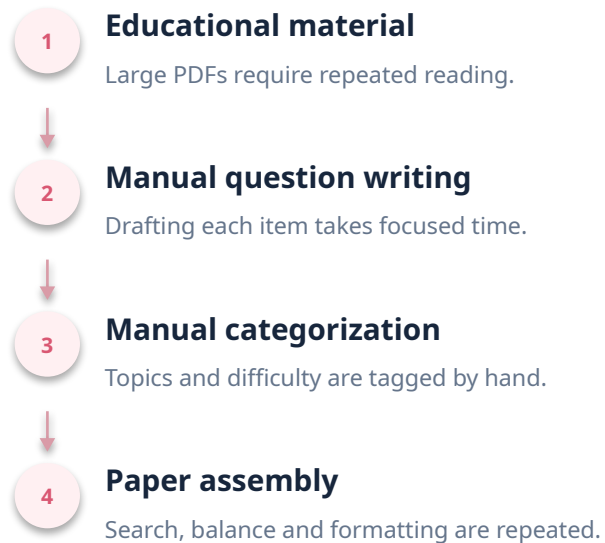
A complete full-stack application — not just a UI prototype.

Why automate question-bank creation?

Manual assessment workflows consume time and make reuse, consistency and discovery harder.

TRADITIONAL

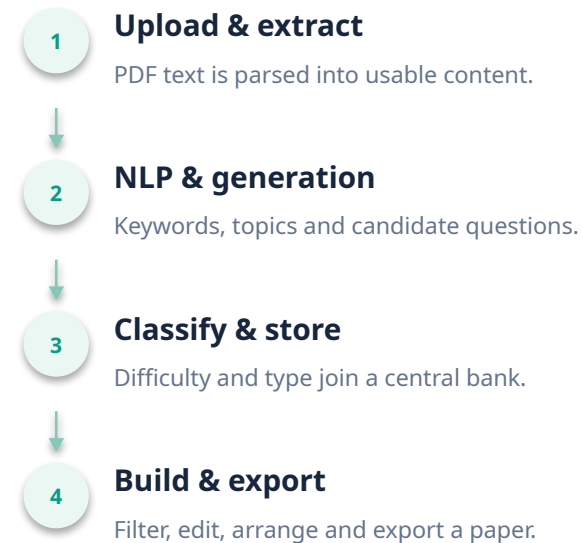
A high-effort, fragmented workflow



PAIN POINTS · time · duplicate risk · scattered records · slow search

PROPOSED

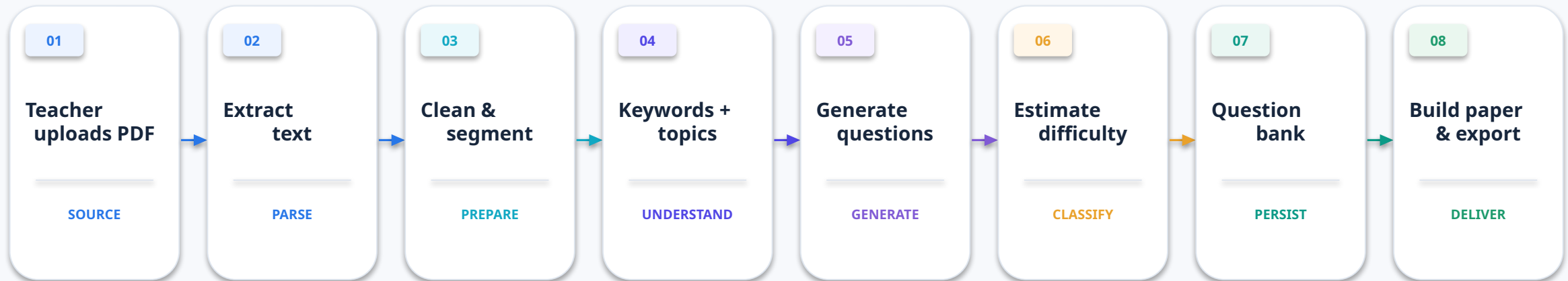
A connected, teacher-controlled system



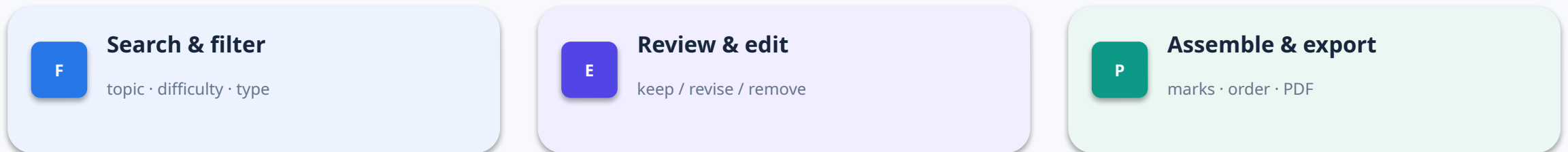
OUTCOME · reusable questions · structured bank · faster paper assembly

Proposed solution: one connected teacher workflow

Educational material enters once; the system turns it into a managed, reusable assessment asset.



AFTER GENERATION · TEACHER REMAINS IN CONTROL



End-to-end path: upload → extract → process → generate → classify → store → curate → publish

Project objectives

The system combines content processing with practical tools for assessment management.

01

Automate question generation

Convert educational content into draft questions.

02

Reduce teacher workload

Cut repeated reading, drafting and formatting.

03

Extract key concepts

Surface important keywords and topics.

04

Classify each question

Assign type and estimated difficulty.

05

Centralize the question bank

Keep material organized and reusable.

06

Reduce duplication

Flag similar questions for review.

07

Enable fast discovery

Search and filter across stored questions.

08

Simplify paper creation

Select, order, score and export to PDF.

Key features

A focused feature set spans document ingestion, question management and paper production.

PDF

Document processing

Upload and process educational PDFs.

PDF · TEXT EXTRACTION

AI

AI / NLP processing

Extract text, keywords and topics.

CLEAN · SEGMENT · IDENTIFY

Q

Multiple question types

MCQ, short, long and true/false.

4 SUPPORTED FORMATS

LVL

Difficulty classification

Estimate easy, medium or hard.

EDITABLE TEACHER REVIEW

FIND

Smart search & filters

Narrow by topic, level, type or source.

FAST QUESTION DISCOVERY

BUILD

Question-paper builder

Select, reorder and assign marks.

CURATED PAPER ASSEMBLY

DATA

Analytics dashboard

View question and topic distributions.

DATABASE-DRIVEN METRICS

PDF

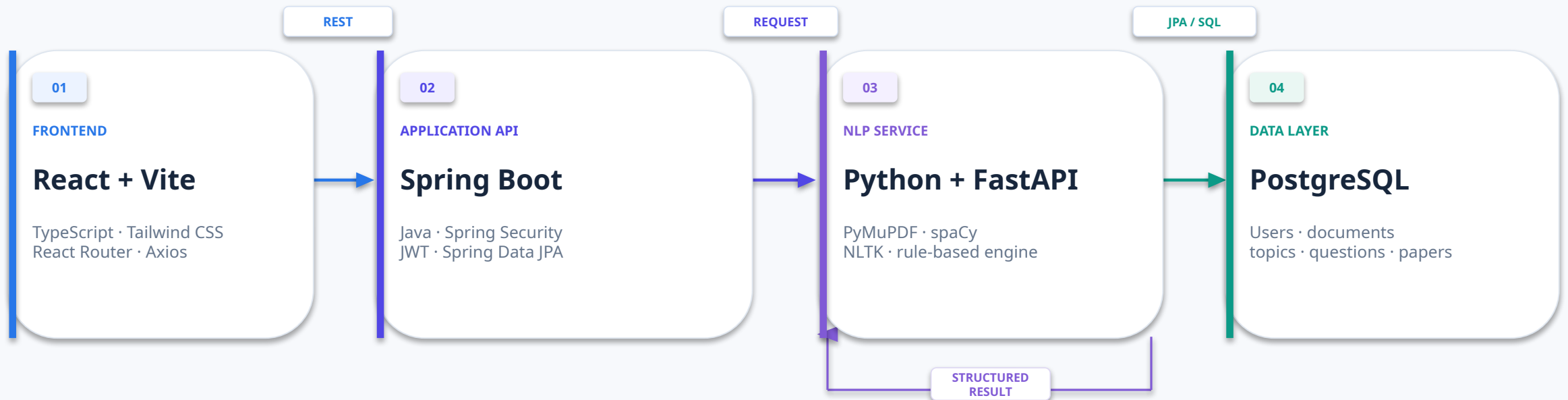
PDF export

Produce a clean, printable paper.

PREVIEW BEFORE DOWNLOAD

System architecture

A modular full-stack design separates the teacher interface, application logic, NLP processing and persistence.



Spring Boot orchestrates the processing request, validates the response and persists reviewed records.

Clear service boundaries support independent testing and future replacement of the local question-generation engine.

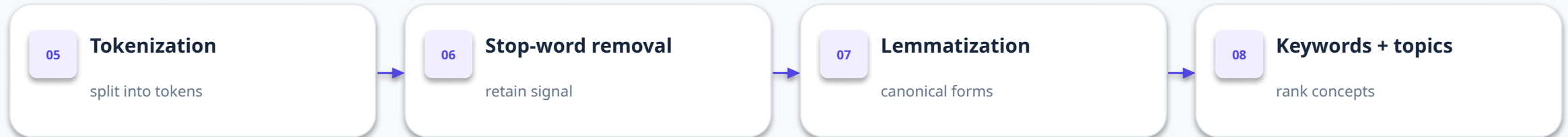
AI / NLP processing pipeline

A transparent sequence turns PDF text into structured concepts, draft questions and an estimated level.

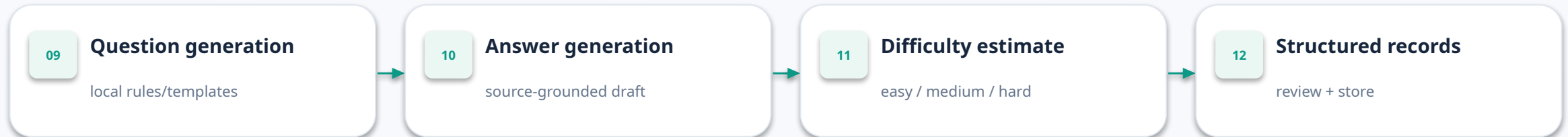
01 · PARSE & NORMALIZE



02 · LINGUISTIC SIGNALS



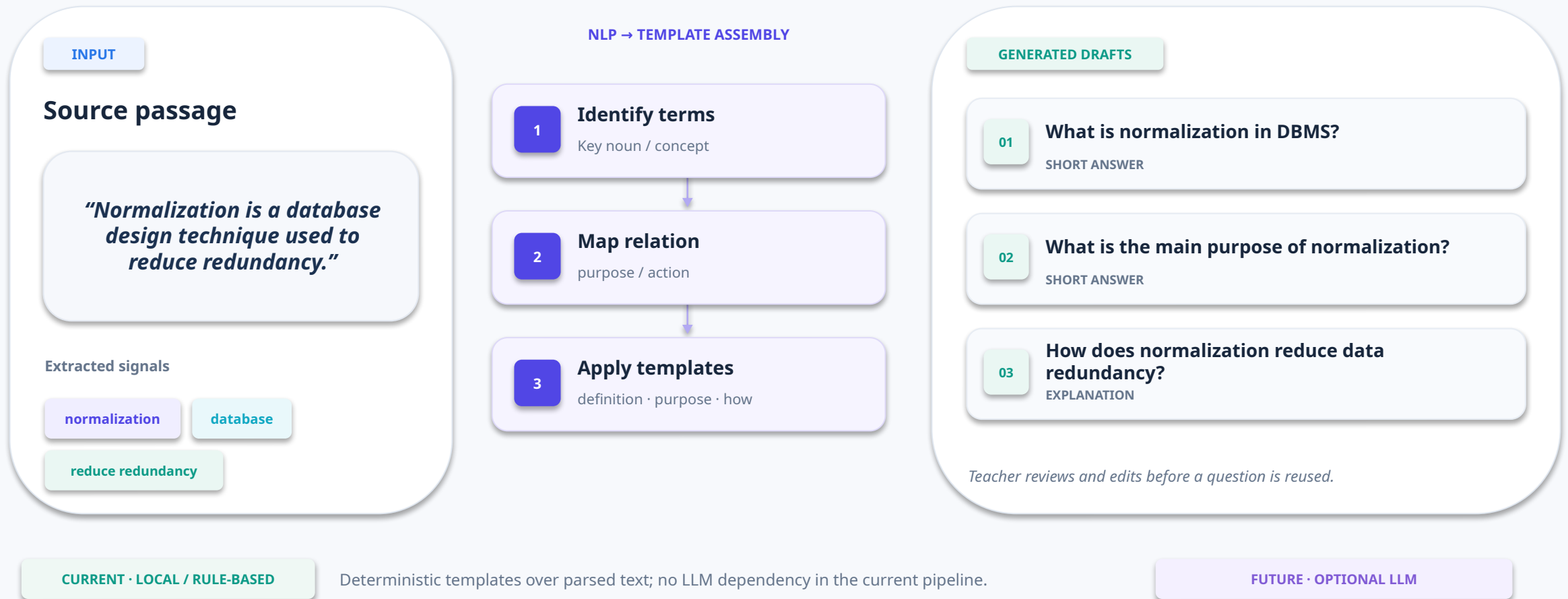
03 · QUESTION & OUTPUT



PROCESSING STACK PyMuPDF → spaCy / NLTK → local rule-based question generation

Question generation engine

Current implementation: rule-based templates operate on extracted concepts; an LLM is an optional future extension.



Question type & difficulty classification

Two independent labels make each item easier to balance, discover and assemble into an assessment.

QUESTION TYPE

MCQ

What is a process?

select one answer

SHORT ANSWER

Define a process.

brief response

LONG ANSWER

Explain process states.

extended response

TRUE / FALSE

A process is a program in execution.

binary response

ESTIMATED DIFFICULTY

1

EASY

Definitions · direct facts

Recall / basic identification

2

MEDIUM

Explain · compare · interpret

Understanding / relation

3

HARD

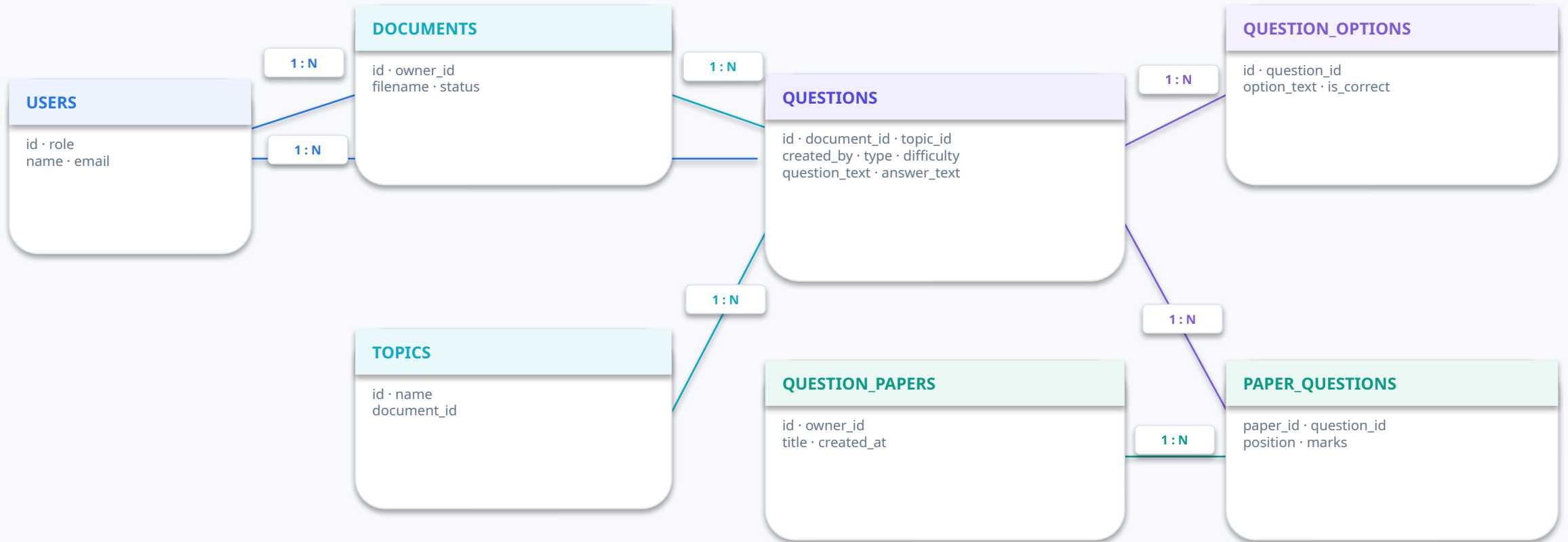
Analyze · apply · solve

Multi-concept reasoning

Classification is an estimate to support teacher review—not a substitute for subject-expert judgment.

PostgreSQL data model

Normalized entities keep ownership, source documents, questions, answer options and assembled papers connected.



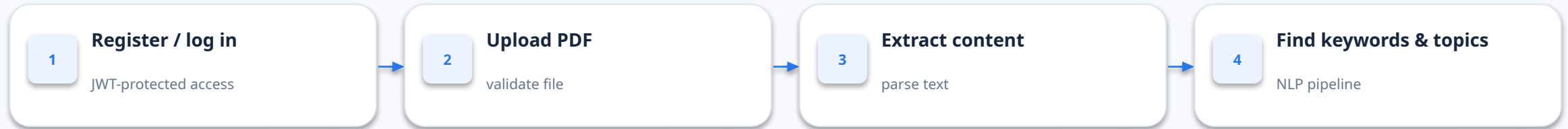
INDEXES

Question text/search · difficulty · type · topic_id · document_id → efficient filtering and retrieval

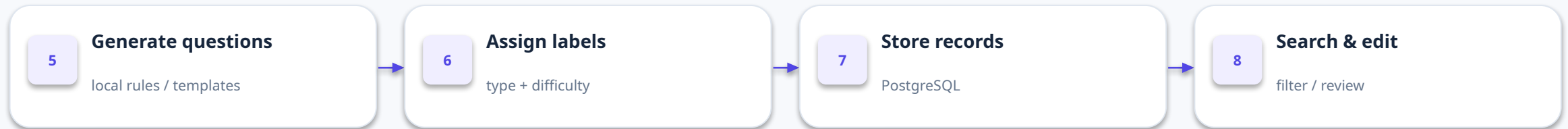
End-to-end application workflow

Twelve steps connect teacher access, document processing, question curation and printable output.

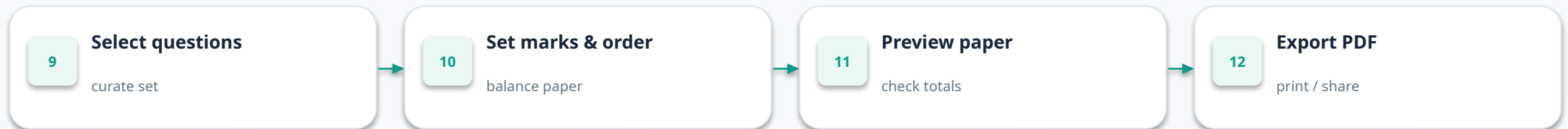
A · INGEST & PROCESS



B · GENERATE & CURATE



C · ASSEMBLE & PUBLISH



INGEST → STRUCTURE → CURATE → PUBLISH

Application interface & main modules

A consistent SaaS-style workspace brings documents, questions, topics and paper building together.

questionbank.app / dashboard

QBank

OVERVIEW

Documents

Question Bank

Topics

Paper Builder

TEACHER

Good morning, Teacher

Your assessment workspace at a glance

142

Questions

08

Documents

12

Topics

Question distribution

Easy

Medium

Hard

Recent documents

DBMS_Unit2.pdf

Processed

OS_Notes.pdf

Ready

Networks.pdf

Queued

DOCUMENTS

Upload document

Drop PDF here or browse

Processing

QUESTIONS

Question bank

Search questions...

What is normalization?

EASY

Explain process states.

MEDIUM

ASSESSMENT

Paper builder

Q1 Define normalization

1 mark

TOTAL 6

Q2 Explain process states

5 marks

PREVIEW

Question-paper generation

Build a balanced paper from curated questions, with marks, order and totals visible before PDF export.

PAPER BUILDER FLOW

1

Select questions

From the searchable question bank

2

Filter the set

Topic · difficulty · question type

3

Assign marks

Per question or by section

4

Reorder the paper

Drag into the desired sequence

5

Validate total marks

Check balance and paper total

6

Preview & export

Review layout, then create PDF

COLLEGE / INSTITUTION NAME

DEPARTMENT OF _____

QUESTION PAPER · DATABASE SYSTEMS

Time: 2 Hours

Maximum: 35 Marks

SECTION A

10 × 1 = 10 marks

1. Define normalization.
2. State one key / constraint.

SECTION B

5 × 2 = 10 marks

11. Explain first normal form.
12. Differentiate key types.

SECTION C

3 × 5 = 15 marks

16. Explain normalization with an example.
17. Analyze an update anomaly.

TOTAL · 10 + 10 + 15 = 35 MARKS

PREVIEW → PDF

Security, reliability & safe handling

Security controls protect teacher accounts, APIs, uploaded files and environment-specific secrets.

DEFENSE-IN-DEPTH CONTROLS

01

AUTHENTICATION

JWT access tokens · BCrypt password hashing

02

AUTHORIZATION

Role-based access · ADMIN / TEACHER

03

API PROTECTION

Input validation · CORS · safe error responses

04

FILE HANDLING

PDF type validation · upload size limits · safe storage

05

CONFIGURATION

Environment variables · DB credentials · JWT secret · AI URL

Reliability practices: appropriate HTTP status codes · consistent API contracts · controlled document processing

TRUST BOUNDARY

Secure by design, clear in operation.

01

Validate early

Reject invalid input at the API boundary.

02

Limit access

Authenticate and authorize each request.

03

Fail safely

Return meaningful status codes without leaking secrets.

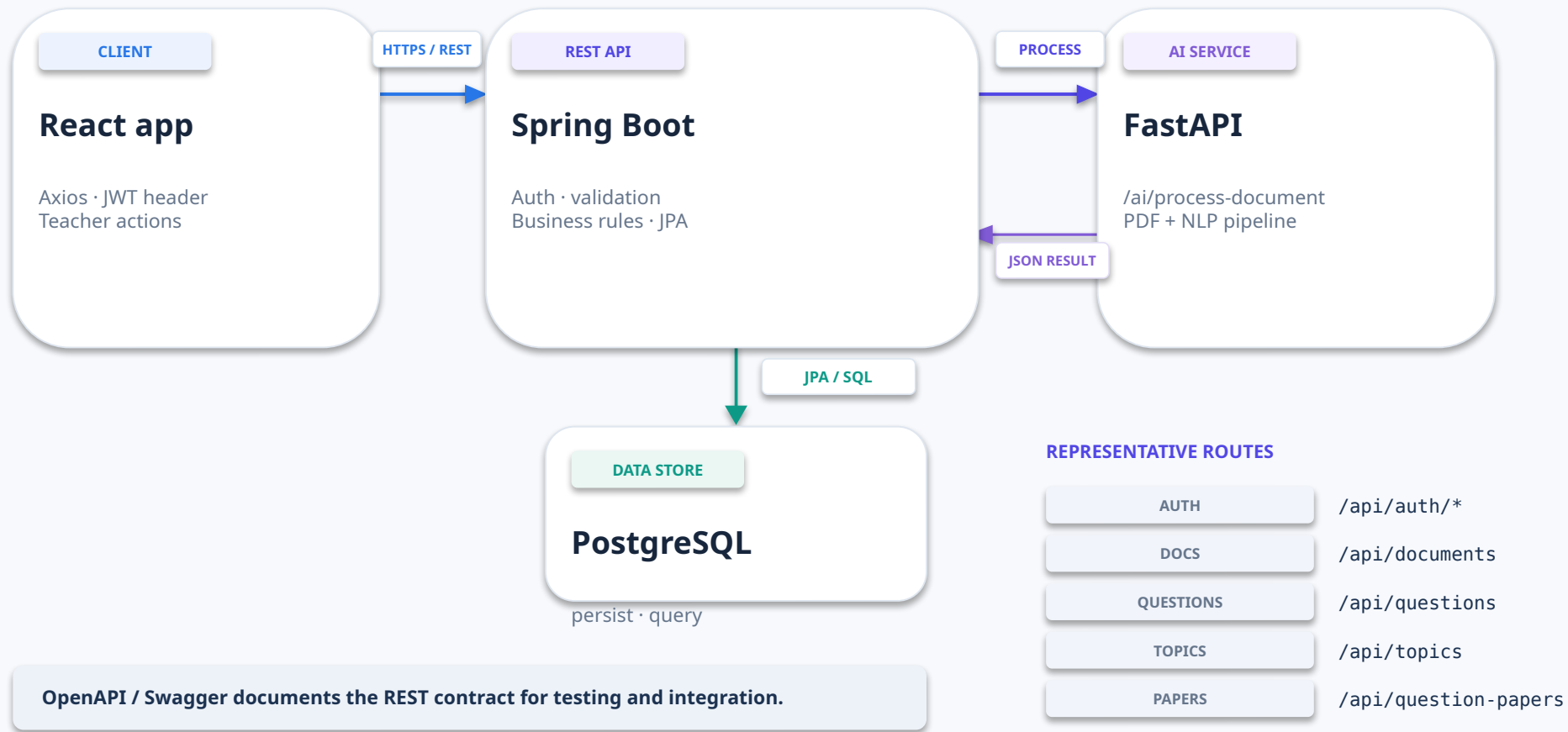
04

Externalize secrets

Keep credentials outside source code.

API & microservice communication

The Spring Boot API is the orchestration boundary between the React client, NLP service and PostgreSQL.



Analytics & teacher dashboard

Database-derived metrics make content coverage and assessment balance visible at a glance.

ILLUSTRATIVE VALUES

142

Questions

08

Documents

58

Easy

55

Medium

29

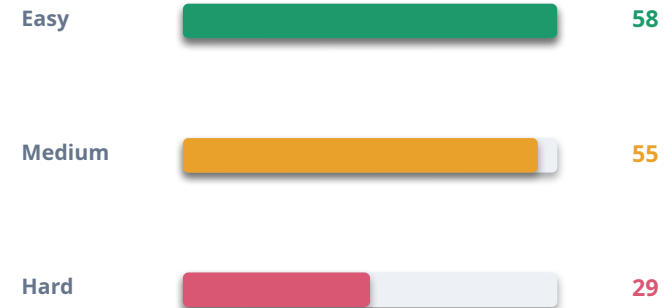
Hard

12

Topics

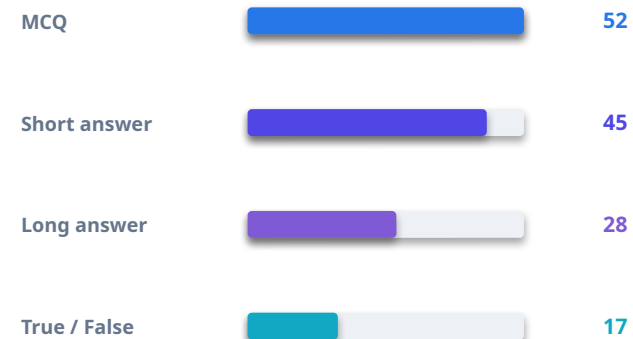
Questions by difficulty

n = 142 questions

 $58 + 55 + 29 = 142$

Questions by type

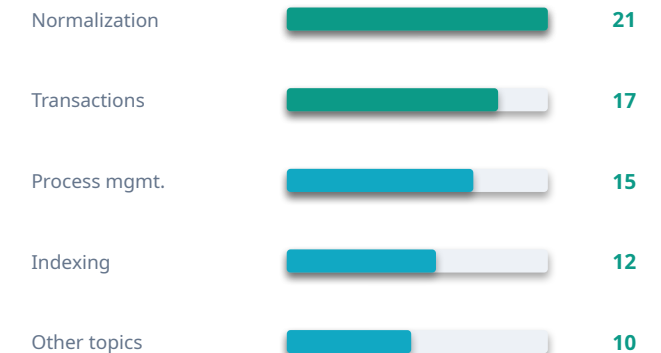
Illustrative distribution



All type totals sum to 142

Top topics by questions

Coverage within the bank

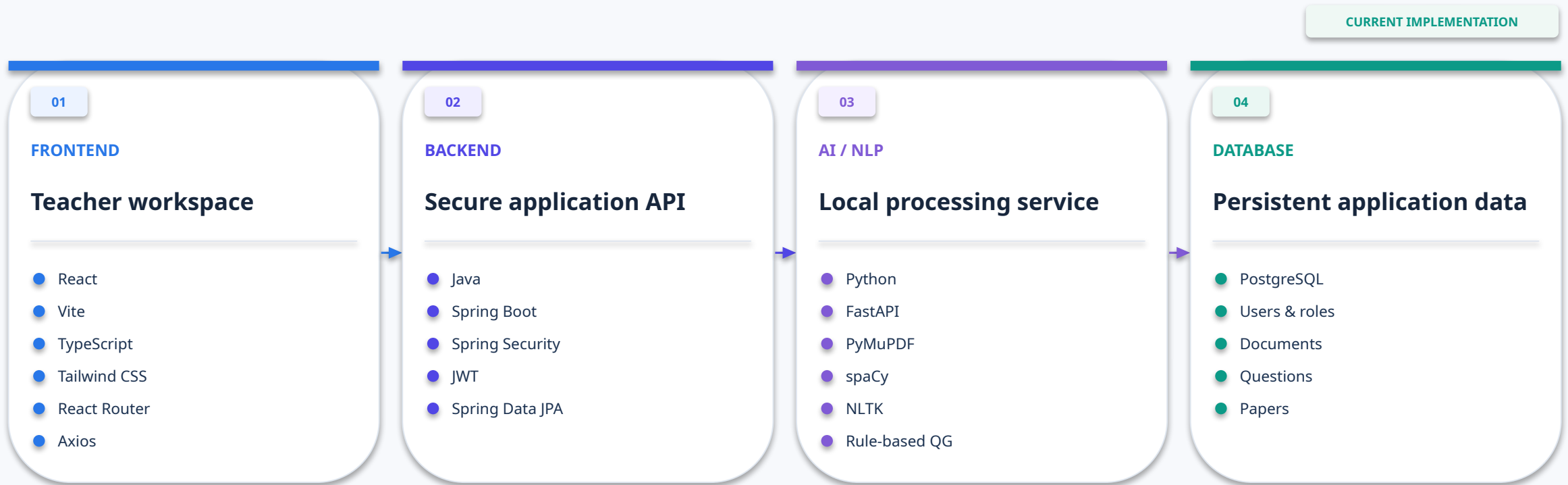


Counts computed from stored records

Dashboard cards and charts are aggregates from PostgreSQL records; values above are an illustrative sample snapshot.

Technology stack & implementation

The implementation uses a practical separation of concerns across web client, REST backend, NLP service and relational data.



IMPLEMENTED PATH

PDF extraction · NLP preprocessing · keywords/topics · rule-based questions · difficulty estimate

REST + OPENAPI

PostgreSQL storage · REST APIs · JWT authentication · question-paper builder · PDF export

DOCKER / COMPOSE

Deployment / packaging: Docker + Docker Compose

Future scope & conclusion

A practical first release today—with clear, achievable extensions for a richer learning-assessment platform.

FUTURE SCOPE

Potential next capabilities

● LLM question generation

● OCR for scanned PDFs

● Multilingual generation

● Question quality scoring

● Bloom's Taxonomy

● ML difficulty prediction

● Cloud document storage

● Email notifications

● Collaborative question banks

CURRENT SYSTEM

From educational PDF to a managed question bank.

The system combines automated content processing with teacher review, searchable storage and a practical paper builder—reducing repeated manual effort.

Thank You

Questions & Discussion

COMPLETE SYSTEM FLOW

